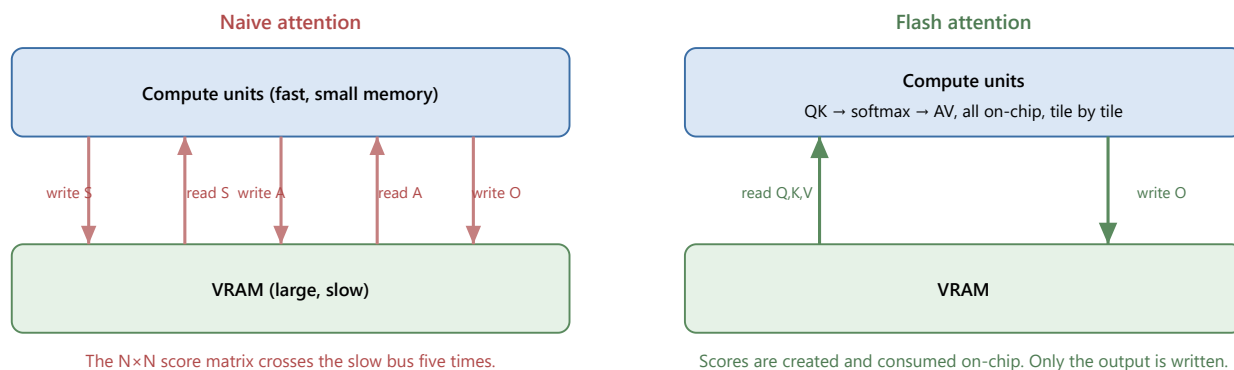




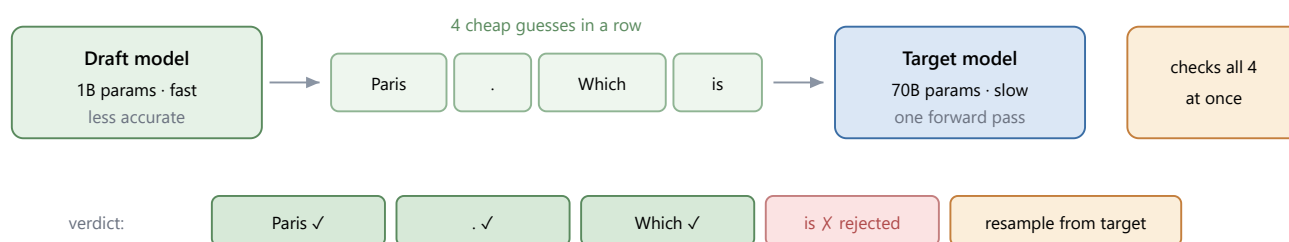
Same arithmetic, same result — far less traffic. Flash attention is mathematically identical to standard attention, not an approximation.

The number of attention scores grows with the *square* of sequence length, so this saving gets more valuable the longer your contexts get. It costs some extra FLOPs during training's backward pass (it recomputes instead of storing), which is a good trade whenever memory traffic is the bottleneck — that is, nearly always. It's standard in every modern inference framework.

Speculative Decoding: A Small Model Drafts, a Big Model Checks

One more observation to exploit. A lot of tokens are boring. After "Dr" comes ".". After "ladies and" comes "gentlemen". Those tokens cost exactly as much as the hard ones — a full 280 GB forward pass each.

So: let a small, fast **draft model** guess the next few tokens. Then have the big **target model** check all of those guesses *in a single forward pass*, because a transformer already predicts the next token at every position at once. Checking is nearly free compared to generating.



Accept rule: if the target model likes the token at least as much as the draft did, accept it outright. Otherwise accept it with probability $p_{\text{target}} / p_{\text{draft}}$ — and if rejected, resample from the corrected distribution.

That rule makes the output mathematically identical to running the big model alone.

Three tokens accepted for the price of one big pass. Rejections cost you nothing but the draft work.

That last point is what makes speculative decoding safe to deploy: it is not an approximation, not a quality/speed trade. Same distribution, fewer big-model passes — typically 50–67% fewer.